Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

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Project Proposal and Objectives

Overall Project Idea: To design a completely decentralized, **Federated Personalized Drift-Aware Attention Framework (FPDAF)** tailored specifically for Multivariate Healthcare Time-Series (e.g., predicting ICU risks like Sepsis from patient vitals).

Objectives to Achieve

1. **Privacy Preservation:** Train predictive models across multiple hospitals collaboratively without sharing raw patient data (Federated Learning).
2. **Personalization:** Use Local Personalization Heads to manage patient heterogeneity (Non-IID distributions).
3. **Drift Adaptation:** Implement CUSUM algorithms to detect when a patient's physiological state degrades and rapidly adapt the model locally.
4. **Clinical Trust:** Provide transparency using Temporal Attention mechanisms so doctors know *why* the model predicted a risk.

Introduction and Background

Introduction to the Problem: In modern Intensive Care Units (ICUs), continuous monitoring systems generate vast amounts of high-frequency multivariate time-series data (Heart Rate, Blood Pressure, SpO2). Detecting critical deterioration early saves lives.

Motivation to Solve This Specific Problem:

- **The Data Silo Problem:** High-accuracy Deep Learning models require massive datasets, but rigid healthcare privacy laws (e.g., HIPAA) forbid combining data into a centralized server.
- **The Global Model Failure:** If multiple hospitals successfully engage in standard Federated Learning (FedAvg), the resulting "global model" fails because hospital demographics and local treatment protocols vary wildly.
- **The Black Box Problem:** Doctors cannot act on an AI alert if the model cannot explain the physiological basis for its warning.

Research Gap Analysis

Existing Research	Identified Limitations / Gap	Need / Why Gap Matters	Project Contribution
Centralized ML and on-device analytics systems (Papers 1, 17, 19, 23)	Do not support federated learning or privacy-preserving collaboration	Centralized data sharing violates privacy laws and is unsuitable for distributed environments	Enables collaborative analytics using federated learning without sharing raw data
Basic Federated Learning frameworks (FedAvg) (Papers 2, 4, 5)	Lack personalization, explainability, and analytics-level insights	A single global model performs poorly on heterogeneous client data	Introduces personalized local models with global aggregation and analytics
Personalized Federated Learning methods (FedPer, pFL) (Papers 3, 10, 16, 20)	Personalization exists but lacks explainability and drift awareness	Without explainability and adaptation, predictions cannot be trusted over time	Combines personalization with explainable analytics and drift handling
Concept driftaware FL approaches (Papers 2, 9)	Drift handling is not integrated with analytics or forecasting pipelines	Real-world data behavior changes over time, degrading prediction accuracy	Integrates drift detection with time-series analytics and forecasting
Federated Time-Series Forecasting works (Papers 6, 8, 1015, 18)	Primarily focus on accuracy; limited explainability and personalization	Stakeholders require transparency to trust and act on predictions	Provides explainable insights alongside federated forecasting
Explainable AI approaches (Papers 7, 17, 19, 23)	Explainability mostly exists in centralized or non-time-series settings	Explainability is critical for trust in sensitive domains like healthcare and energy	Adapts explainable AI techniques to federated time-series analytics
Healthcare-focused FL applications (Papers 9, 21, 22, 24, 25)	Application-specific solutions with limited generalization	A reusable framework is required across multiple domains	Proposes a domain-agnostic federated analytics framework
Security- and privacy-enhanced FL methods (Papers 13, 24, 25)	High complexity and low usability despite strong privacy guarantees	Practical systems must balance privacy, performance, and usability	Balances privacy, efficiency, and interpretability in a single framework

Dataset: PhysioNet Challenge 2019 (Sepsis Prediction)

Dataset Overview:

- **Source:** PhysioNet (Public Healthcare Dataset).
- **Data Type:** Real-world multivariate ICU patient monitoring time-series.
- **Structure:** 40,000 patient files (`.psv`), with each row representing one hourly observation.
- **Key Features:** Heart Rate, BP, SpO2, Temperature, ICULOS.
- **Target:** `SepsisLabel` (predicting deterioration).

Why PhysioNet?

- **Heterogeneity:** `setA` & `setB` come from different hospitals (perfect for FL).
- **Time-Series:** Sequences support LSTM & Drift tracking.
- **Clinical:** Interpretable vitals support our XAI mechanism.

Execution Plan (Components & Flow)

Subdomain	Matching Components Identified	Execution Target
Data Engineering	PhysioNet Sepsis Database, Time-series normalization, Sequential chunking.	Preprocessing Pipeline
Deep Learning Base	LSTM/Transformer backbone, Temporal Attention Layer.	Global Model
Decentralization	Federated Averaging (FedAvg) algorithm, Secure Gradient Aggregation.	Federated Loop
Customization	Local Personalization Heads (*FedPer*).	Client Local Model
Statistical Math	Cumulative Sum (CUSUM) for tracking prediction residual drift.	Drift Tracker Module

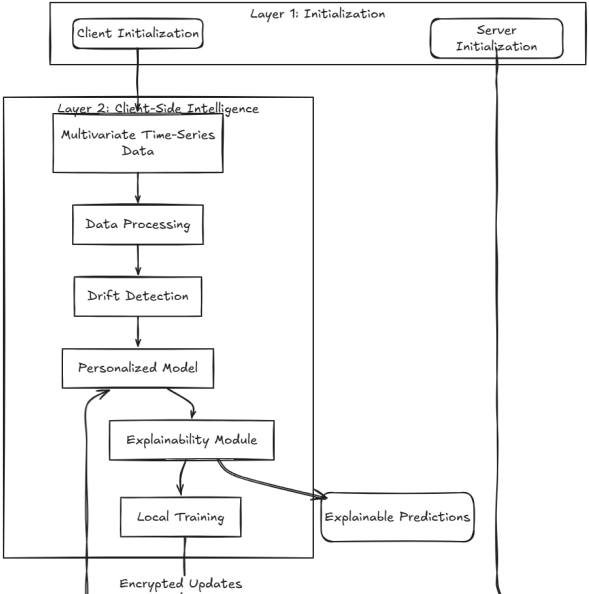
Solution Pipeline: Extract Data $\rightarrow$ Train Local Models $\rightarrow$ Identify Attention Metrics $\rightarrow$ Sync Shared Layers Globally $\rightarrow$ Monitor Sub-populations for Drift $\rightarrow$ Adapt Local Heads.

Fundamentals Learned in the Semester

To execute the plan above, our team embarked on extensive upskilling over this semester:

- **Federated Learning Architecture:** Extensively reviewed *McMahan et al.'s* fundamental papers to understand gradient averaging and the shift from Centralized ML to Decentralized ML.
- **PyTorch Framework Fundamentals:** Completed MOOCs & online tutorials to transition our team's technical skills into PyTorch (specifically custom `nn.Module` networks, `DataLoaders`, and customized training loops).
- **Time-Series Analytics:** Studied LSTM sequence modeling and Attention-mechanism math (computing context vectors via softmax weights).
- **Statistical Process Control:** Learned the *CUSUM* (Cumulative Sum) statistical methodology for detecting small shifts in ongoing time-series data.

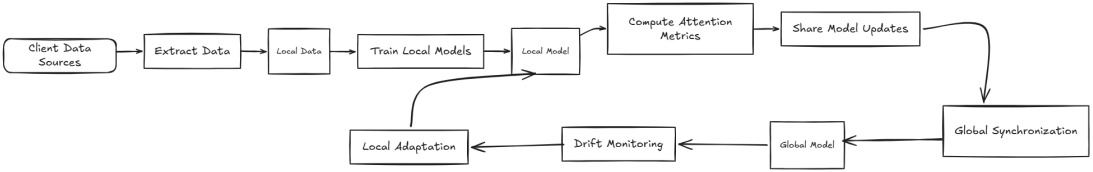
Architecture Diagram and Explanation of Design



Why this design?

- **Privacy:** Only encrypted gradients travel upward.
- **Local LSTM Base:** Captures generalized physiological rules.
- **Attention Layer:** Provides XAI (feature importance) right before the local prediction.

System Dataflow Mapping



Summary of Works Completed (Technical Evidence)

Phase 1 Contributions mapping to Technical Validation & Evidence:

1. **Problem Formulation:** Successfully scoped the application domain specifically to healthcare ICU telemetry.
2. **Mathematical Formulation:** Defined formulas for exactly how CUSUM and Attention will be mathematically integrated into the *FedPer* pipeline.
3. **Architecture Design:** Finalized the multi-layered system diagram.
4. **Repository Framework Setup (Baseline Code):**
   - Initialized project GitHub repository.
   - Pulled and set up the foundational **FedAvg-PyTorch base framework**.
   - Built initial custom PyTorch stubs simulating the isolated ‘LocalPersonalizedModel‘ and a synthetic ‘SimulatedICUDataset‘ proxy.

Plans for the Oncoming Semesters (Phase 2)

The direct roadmap to full implementation:

Target Area	**Objective for Phase 2**
Dataset Setup	Download and process the full PhysioNet 2019 ICU Time-Series database.
Baseline Benchmarking	Train standard *FedAvg* on PyTorch to establish RMSE/MAE baselines.
Sub-Model Integration	Code the *CUSUM* mathematical trigger and the *Attention* weight visualizer.
Simulation Runs	Simulate 3-5 distinct Hospital "Clients" locally to monitor Federated converg
Final Evaluation	Quantify percentage gains in accuracy and personalization against baselines.

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Thank You Questions?